

Probability

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Chapter 1

Theory

1.0.0.1 Matrix Match Type Problems

Matrix 1: A student goes to exam on a bicycle or scooter or car or on foot. The probabilities of his using bicycle, scooter car or being on foot are $\frac{1}{10}$, $\frac{1}{5}$, $\frac{3}{10}$ or $\frac{2}{5}$ respectively. The probability of reaching late at the examination centre by using these modes are $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{12}$ and $\frac{1}{24}$ respectively. The student reached the exam centre on time. Under Column I, the modes of travel are given and under Column II, the probabilities that the student used the particular mode of transport provided that he reached the exam centre on time. Match the corresponding entries.

Column I	Column II
(P) Bicycle	(A) $\frac{2}{5}$
(Q) Scooter	(B) $\frac{1}{5}$
(R) Car	(C) $\frac{2}{15}$
(S) On Foot	(D) $\frac{4}{15}$

1.0.0.2 Linked Comprehension Type Questions

Comprehension 1: There are n biased coins and n boxes numbered 1 to n . The coins are tossed and each put into its corresponding box. For the k^{th} coin, the odds in favour of appearance of head are k and the odds against are $2n - k$. Let E_n denote the event of selecting the k^{th} box and H denote the event that the selected box contains a Head.

Q1: If $P(E_k) = c \forall k$, then $P(E_m|H)$ where $m \in \{1, 2, 3, \dots, n\}$ is

- A) $\frac{m}{n(n+1)}$
- B) $\frac{2m}{n(n+1)}$
- C) $\frac{c}{n+1}$

D) $\frac{2}{n+1}$

Q2: If $P(E_k) \propto k$ for $k = 1, 2, 3, \dots, n$. Find $P(H)$

A) $\frac{2n+1}{6n}$

B) $\frac{2n+1}{4n}$

C) $\frac{2}{3}$

D) $\frac{1}{3}$

Q3: Evaluate the probability P that only the m^{th} box contains a head. Its value for $m = 3, n = 5$ is

A) 0.12

B) 0.0832

C) 0.648

D) None of these

1.0.0.3 Hints and Solutions

Matrix Match Type Problems

Matrix 1: Answers

P	A	B	C	D	E
Q	A	B	C	D	E
R	A	B	C	D	E
S	A	B	C	D	E

Linked Comprehension Type Questions

Comprehension 1 Answers Q1) B Q2) A Q3) C

Chapter 2

Advanced

Q1: Three students S_1, S_2 , and S_3 are given a problem to solve. Consider the following events:

U : At least one of S_1, S_2 , and S_3 can solve the problem,

V : S_1 can solve the problem, given that neither S_2 nor S_3 can solve the problem,

W : S_2 can solve the problem and S_3 cannot solve the problem,

T : S_3 can solve the problem.

For any event E , let $P(E)$ denote the probability of E . If

$P(U) = \frac{1}{2}$, $P(V) = \frac{1}{10}$, and $P(W) = \frac{1}{12}$.

then $P(T)$ is equal to :

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(A) $\frac{13}{36}$

(B) $\frac{1}{3}$

(C) $\frac{19}{60}$

(D) $\frac{1}{4}$

{ Solution : <https://www.youtube.com/watch?v=gjzClhgArM4> }